thermal-bond 3.0F VT-4B3H RCF

UL Approval: E214381 Version: 14/10/2026

Thermally Conductive Resin Coated Film Bondply

Resin Coated Film (RCF) Bondply is an unreinforced adhesive system coated onto PET film for use in high performance and high reliability multilayer PCB stack-ups.

VT-4B3H is a high Tg, ceramic-filled, thermally conductive (3.2 W/mK), halogen-free thermoset resin system, specifically designed for use in multilayer PCBs requiring enhanced levels of heat management and thermal conductivity. Thermal-bond RCF can be combined with other Ventec laminate & prepreg systems including tec-thermal IMS, VT-5A2 and VT-4A2H thermally conductive laminates & prepregs in hybrid stackups.

General Information

- Thermal Conductivity – 3.2 W/mK
- Tg 180ºC
- Ceramic Filled
- Halogen Free
- Flammability (UL94 V-0)
- UL MOT 155ºC
- Easy handling (like rubber)

Application

- High Thermal
- High Working Voltage [≥500Volts]
- Heavy Copper Filled
- Power Conversion
- Monitor Drives
- Rectifiers, Power Supply
- Metal in Board (MiB) applications including Coins and Inserts, Pedestal, etc.

Availability

Press Ply Thickness	75µ/100µ/125µ/150µ (0.003"/0.004"/ 0.005"/ 0.006")
Panel Size	460*610mm, 510*610mm, 533*610mm. 18.11*24.02", 20.08*24.02", 20.98*24.02". or as required

Carrier Film Type	
PET (Standard)	T

RCF Part Numbers

Description	Part Number	PPT (µm)	
thermal-bond 3.0F RCF Medium Flow	4B3H-FTM-075	75	Multilayer MCPCB. High copper retention, normal copper and hole filling requirements.
	4B3H-FTM-100	100	
	4B3H-FTM-125	125	
	4B3H-FTM-150	150	
thermal-bond 3.0F RCF Regular Flow	4B3H-FTR-075	75	Normal multilayer MCPCB. Low-thickness copper foil filling, no need for hole filling & Cu pedestal.
	4B3H-FTR-100	100	
	4B3H-FTR-125	125	
	4B3H-FTR-150	150	
thermal-bond 3.0F RCF High Flow	4B3H-FTH-075	75	Multilayer MCPCB, resin hole filling, embedded PCB with high-power components. Low copper retention, heavy copper construction (>3oz), stringent requirements for hole filling
	4B3H-FTH-100	100	
	4B3H-FTH-125	125	
	4B3H-FTH-150	150	

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Properties

Properties		Test Method	Units	75µm	100µm	125µm	150µm
Thermal Properties							
Thermal Conductivity		ISO 22007-2	W/mK	3.2			
Thermal Impedance		ISO 22007-2	°C*in²/W	0.036	0.048	0.061	0.072
Tg	DMA	IPC-TM-650 2.4.24.4	°C	180			
Td	TGA	ASTM D3850	°C	395			
Thermal Stress @ 288°C solder dip		IPC-TM-650 2.4.13.1	Minute	≥5			
X/Y CTE	<Tg	IPC-TM-650 2.4.24.5	ppm/°C	17			
Z-axis CTE	Total Expansion (50~260°C)	IPC-TM-650 2.4.24	%	1.9			
	Before Tg	IPC-TM-650 2.4.24	ppm/°C	40			
	After Tg	IPC-TM-650 2.4.24	ppm/°C	85~90			
Electrical Properties							
Dk @ 1MHz	C-24 / 23 / 50	IPC-TM-650 2.5.5.3	–	4.8			
Df @ 1MHz	C-24 / 23 / 50	IPC-TM-650 2.5.5.3	–	0.016			
Volume Resistance	After Moisture Resistance	IPC-TM-650 2.5.17.1	MΩ-cm	5.0E+8			
	E-24/125	IPC-TM-650 2.5.17.1	MΩ-cm	3.0E+7			
Surface Resistance	After Moisture Resistance	IPC-TM-650 2.5.17.1	MΩ	2.0E+7			
	E-24/125	IPC-TM-650 2.5.17.1	MΩ	5.0E+6			
Hi-Pot Withstand	DC	IPC-TM-650 2.5.7	Volt	2000	3000	4000	5000
Breakdown Voltage	AC	IPC-TM-650 2.5.6.3	Volt	7000	8000	9000	10000
Mechanical Properties							
Peel Strength (1oz)	As received	IPC-TM-650 2.4.8	lb/in	8.0			
CTI	As received	ASTM D3638	Volt	600			
Physical Properties							
Flammability	As received	UL-94	Rating	V-0			

Note: All test data provided are typical values and not intended to be specification values.

Storage Condition

		RCF	
Storage Condition	Temperature	< 23°C (73°F)	< 5°C (41°F)
	Relative Humidity	< 55%	/

Disclaimer: The information and data contained in this technical literature is based on data and knowledge correct at the time of publishing/printing and is believed to be accurate and is offered in good faith for the benefit of the user. The user should make his own tests to verify the suitability of this product for any application before its use. All data are typical values only and subject to change without notice.